

**FIBROTEK***Precision in Every Spindle***KEY FEATURES****Hairiness CV Control****1**

Continuously regulates suction airflow at the compacting zone to control Yarn Hairiness Index and Hairiness CV%. By ensuring uniform integration of protruding fibers into the yarn body, the system maintains a consistent, low-hairiness yarn surface across all spindles and throughout the doff. Reduced hairiness (H, S3 value) directly improves fabric surface appearance, reduces pilling tendency, and minimizes downstream processing issues such as size-box and warping fly generation.

**Yarn Strength Enhancement****2**

Improves fiber-to-fiber cohesion within the compact zone through optimized and stable suction control, resulting in better fiber alignment and parallelization during yarn formation. This translates into higher yarn tenacity (cN/tex) and RKm, improved evenness (U%), and reduced Imperfections (IPI — thin places, thick places, and neps). The result is fewer end-breaks at the ring frame and in downstream winding, warping, and weaving/knitting operations.

**Automatic Shutter Duct Cleaning****3**

Performs an automatic high-suction cleaning cycle of the shutter duct at the end of every machine doff. Accumulated fiber fly, lint, and dust are cleared from the suction duct without manual intervention, keeping suction airflow uniform across all spindle positions. This prevents duct choking, avoids localized loss of suction that leads to hairiness variation, and reduces the manual cleaning workload for machine operators.

**Low Suction Pressure Detection & Alarm****4**

Continuously monitors suction (negative pressure) in the duct system and instantly detects any drop below the required operating threshold. On detection of low suction pressure caused by duct blockage, filter choking, or fan wear the system raises an immediate alarm, allowing operators to intervene before yarn quality is affected. This protects against inconsistent hairiness control and helps plan preventive maintenance before a quality deviation occurs.

**Compatible Pressure Transmitter****5**

FIBROTEK can be supplied with a compatible, factory-calibrated industrial pressure transmitter matched to the suction duct system. The transmitter provides accurate, real-time negative-pressure feedback to the controller, enabling precise closed-loop suction control rather than fixed open-loop operation. This ensures the suction level automatically tracks actual duct conditions for consistent hairiness performance batch after batch.

**Energy-Efficient Inverter Control****6**

Drives the suction blower motor through an optimized Variable Frequency Drive (VFD) control strategy that adjusts fan speed according to actual suction demand, instead of running the motor continuously at fixed full speed. This demand-based control reduces electrical power consumption, lowers blower and motor wear, cuts operating noise, and extends the service life of the suction system while still maintaining the suction pressure required for stable hairiness control.

**Why choose FIBROTEK?**

- ✓ Reduced Yarn Hairiness (H & CVh%)
- ✓ Improved Yarn Tenacity (cN/tex & RKm)
- ✓ Better Evenness (U%) & Fewer Imperfections
- ✓ Consistent Compact Yarn Quality Across Spindles
- ✓ Automatic Shutter Duct Cleaning at Every Doff
- ✓ Reduced Manual Cleaning & Downtime
- ✓ Real-Time Suction Pressure Monitoring
- ✓ Instant Low Suction Pressure Alarm
- ✓ Optional Factory-Calibrated Pressure Transmitter
- ✓ Energy-Efficient VFD-Based Suction Control
- ✓ Reduced Power Consumption & Fan Noise
- ✓ Extended Blower & Motor Service Life